Llama 3.2

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Meta Llama 3.2 is a series of advanced open-source large-scale language models (LLMs) developed by the Meta AI department.

1. Model Size

Model	Size
llama3.2:1b	1.3GB
llama3.2:3b	2.0GB

The Raspberry Pi 5 2GB version can only run models with 1B parameters.

2. Performance

Lightweight instruction-tuned benchmarks

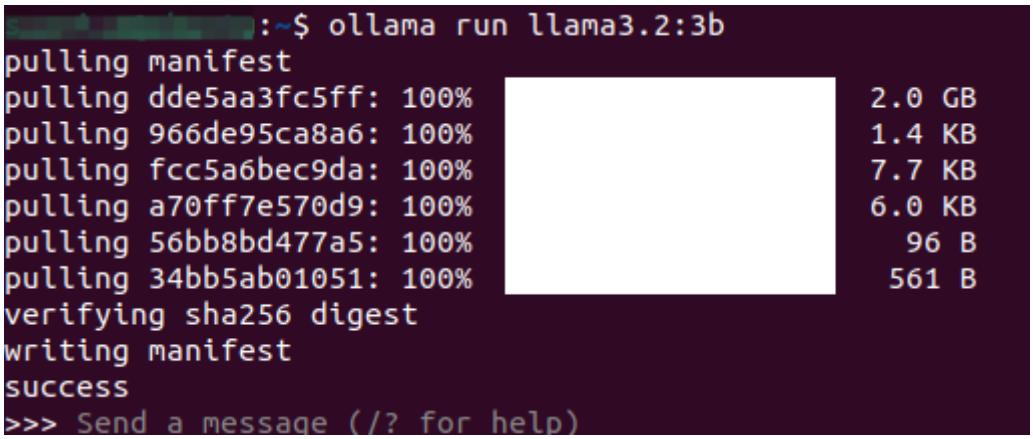
Category Benchmark	Llama 3.2 1B	Llama 3.2 3B	Gemma 2 2B IT (measured)	Phi-3.5-mini IT (measured)
General MMLU (5-shot)	49.3	63.4	57.8	**69.0**
Open-rewrite eval (0-shot, rougeL)	**41.6**	40.1	31.2	34.5
TLDR9+ (test, 1-shot, rougeL)	16.8	**19.0**	13.9	12.8
IFEval	59.5	**77.4**	61.9	59.2
Tool Use BFCL V2	25.7	**67.0**	27.4	58.4
Nexus	13.5	**34.3**	21.0	26.1
Math GSM8K (8-shot, CoT)	44.4	77.7	62.5	**86.2**
MATH (0-shot, CoT)	30.6	**48.0**	23.8	44.2
Reasoning ARC Challenge (0-shot)	59.4	78.6	76.7	**87.4**
GPQA (0-shot)	27.2	**32.8**	27.5	31.9
Hellaswag (0-shot)	41.2	69.8	61.1	**81.4**
Long Context InfiniteBench/En.MC (128k)	38.0	63.3	—	39.2
InfiniteBench/En.QA (128k)	20.3	19.8	—	11.3
NIH/Multi-needle	75.0	84.7	—	52.7
Multilingual MGSM (0-shot, CoT)	24.5	**58.2**	40.2	49.8

3. Using Llama 3.2

3.1 Running Llama 3.2

Use the run command to start running the model. If the model is not already downloaded, it will automatically pull the model from the Olama model library:

```
ollama run llama3.2:3b
```



3.2 Starting a Conversation

```
How many minutes are there in a day?
```

Response time depends on your hardware configuration, so please be patient!

```
>>> How many minutes are there in a day?
There are 1440 minutes in a day. This is calculated by multiplying the
number of hours in a day (24) by the number of minutes in an hour (60).

24 hours/day * 60 minutes/hour = 1440 minutes/day

>>> Send a message (/? for help)
```

3.3 Ending a Conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

3.4 Chinese Conversation

If you don't have a Chinese input method, please refer to the Chinese input method switching tutorial

Chinese Conversation:

```
:~$ ollama run llama3.2:3b
>>> 一个小时是多少分钟，请用中文告诉我。
一个小时有60分钟。

>>> Send a message (/? for help)
```

References

> **Ollama**

Official website:https://ollama.com/

GitHub：https://github.com/ollama/ollama

> **Llama 3.2**

Official website:https://www.llama.com/docs/model-cards-and-prompt-formats/llama3_2/

Ollama corresponding model:https://ollama.com/library/llama3.2